

- **Problem Statement ID – 26080**
- **Problem Statement Title – Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts**
- **Theme – Smart Automation**
- **PS Category – Software**
- **Team ID-**
- **Team Name (Registered on portal)**

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RegimeCast

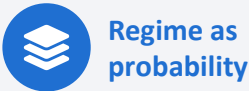
Regime-aware AI post-processing that corrects NWP rainfall forecasts — it never replaces the weather model

Proposed Solution

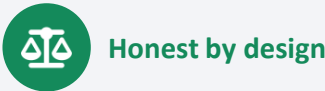
“The same forecast — corrected for the monsoon it is actually in.”

- **Regime Detection**: reads the forecast’s own **850 hPa wind, pressure & moisture** → Active / Break / Depression / Normal, plus orographic & coastal zones
- **Regime-Aware Correction**: removes the NWP model’s bias with the **regime as an input** (LightGBM + quantile mapping)
- **Heavy-Rain Probability**: **calibrated** chance of ≥ 64.5 mm and ≥ 115.6 mm — IMD’s own warning thresholds
- **District Product**: area-weighted onto **all 641 districts** (Survey of India based boundaries)
- **Verified, not claimed**: RMSE · ETS · CSI · POD · FAR · FSS with **significance tests** on every improvement

Why it is different



The corrector sees the classifier’s uncertainty, not a hard guess

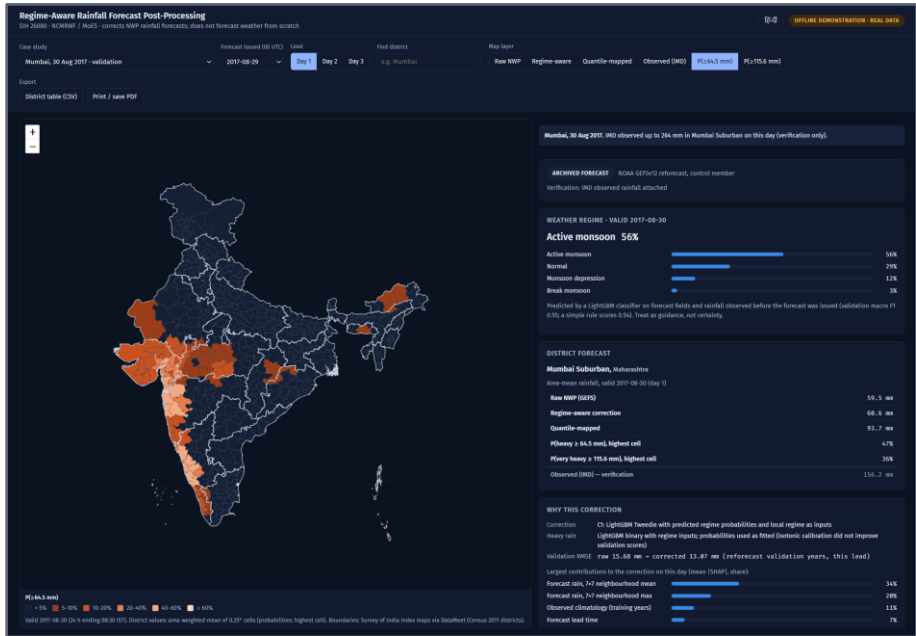


Block-bootstrap tests; reports where regimes do NOT help



Every forecast ships its own SHAP feature contributions

How it works



Working prototype · real archived data · Mumbai deluge, 30-Aug-2017 (validation year)

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TECHNICAL APPROACH

Technology stack & system architecture — built and running end-to-end on real data



Technology Stack

Layer	Tools
Data & geo	xarray · NetCDF / GRIB (ecCodes) · GeoPandas · Shapely
ML / AI	LightGBM · PyTorch U-Net corrector · CNN classifier (tested)
Regimes	Rajeevan (2010) active/break · IMD best-track · terrain forced ascent
Verification	RMSE · ETS · CSI · POD · FAR · FSS · Brier · bootstrap CIs
Backend	Python 3.12 · FastAPI · Pydantic
Frontend	React · TypeScript · Leaflet (fully offline maps)
Quality	179 automated tests (pytest + Vitest)
Open data	NOAA GFSv12 (AWS) · IMD 0.25° rainfall · IMD best-track

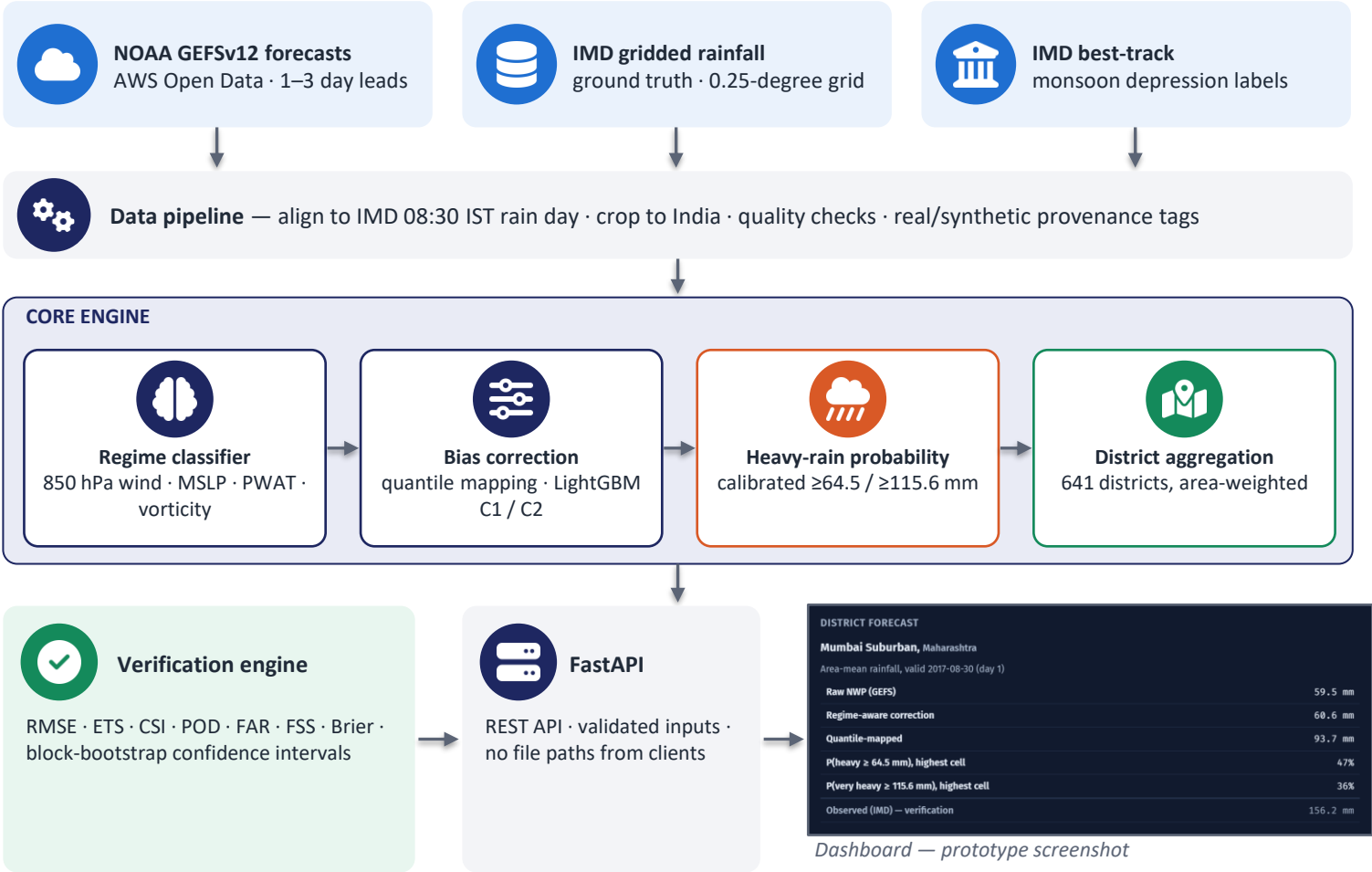


✓ Observations never used as model inputs

✓ Held-out years scored once

✓ Real vs synthetic data tagged and never mixed

System Architecture Flow



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FEASIBILITY AND VIABILITY



✓ FEASIBILITY — already proven, not projected

18

monsoon seasons of real data trained & validated (2000–17)

29.1 M

training rows (forecast ↔ observation)

641

districts covered

179

automated tests passing

- **Runs on a laptop CPU:** forecast product in **~10 s**, full retrain **~1.5 hours** — no GPU
- **Open data, open source:** NOAA GEFS (AWS) + IMD data — zero licence cost
- **Live mode ✓**: today's NOAA GEFS run → district forecast in **< 1 min**

RISKS → HOW WE HANDLE THEM



NCMRWF's NCUM forecast archive is not public



Validated on NOAA GEFS as proxy; adapter retrains on NCUM once shared



Rare regimes (depressions) have few training days



Trained on 16 seasons (2000–15); pooled models with regime inputs + fallback



Regime classification is imperfect (macro-F1 0.55)



Regime passed as probabilities; transparent rule baseline; evidence tags on every rule

🚀 VIABILITY — path to operations



① **Prototype ✓**

Validated 2016–17 + one-time held-out test (real data)



② **Live mode ✓**

Tested on 610 real NOAA daily forecasts (2021–25): AUC 0.88



④ **Daily operations**

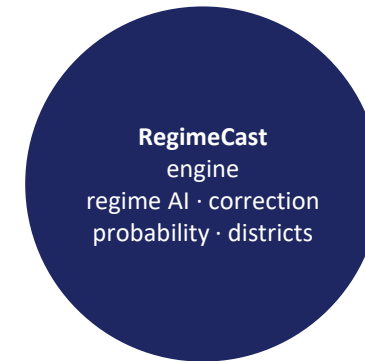
Products every forecast cycle → IMD / NCMRWF dashboards



③ **Plug in NCMRWF data**

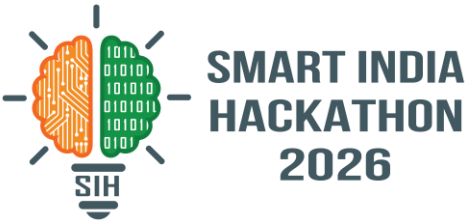
Retrain on NCUM / NEPS via the ready adapter; re-verify

continuous
re-verification



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IMPACT AND BENEFITS



Measured impact — unseen real data, scored once

0.56 → 0.90

heavy-rain detection (ROC AUC), test years 2018–19: raw forecast vs our model

–20%

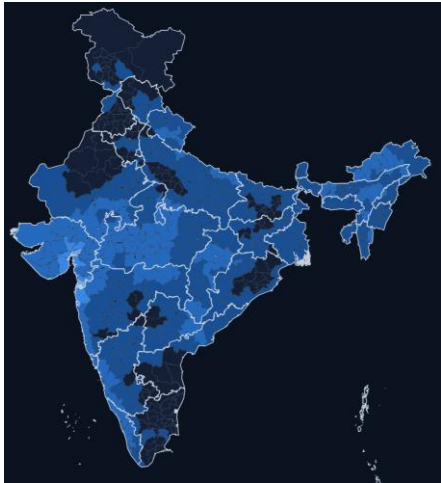
day-1 forecast error (RMSE), test years 2018–19; –12% on 610 live NOAA forecasts

47%

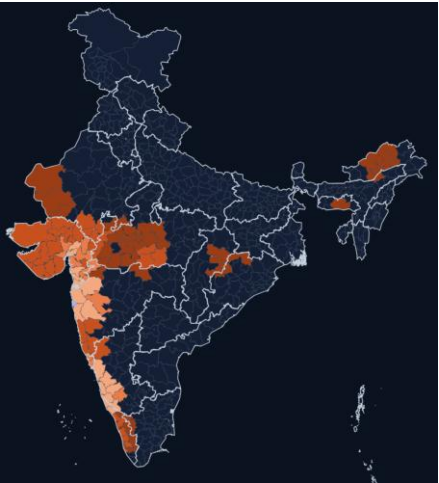
heavy-rain chance flagged for Mumbai Suburban vs 1.4% base rate

Case study · Mumbai deluge — rain day ending 08:30 IST, 30-Aug-2017, forecast issued 29-Aug (day 1)

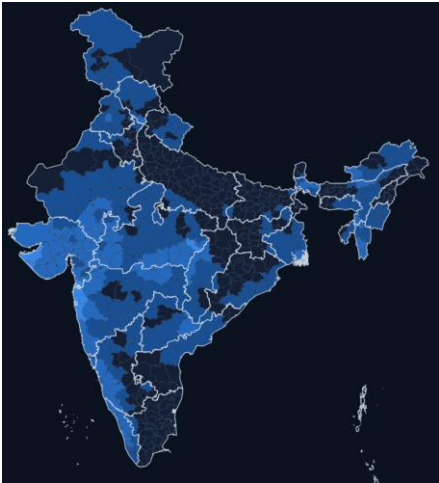
Raw GEFS forecast (mm)



Our P(heavy rain ≥64.5 mm)



IMD observed (mm)



Mumbai Suburban: raw 60 mm · regime-aware 61 mm · U-Net 67 mm · quantile-mapped 94 mm · observed 156 mm — amounts under-forecast the extreme, the **probability layer flags it**
Validation-year case (2017). Dark-mode maps: brighter = more rain (blue) or higher probability (orange)

Benefits of the solution

SOCIAL

Earlier district-level heavy-rain warnings for flood preparedness

ECONOMIC

Better sowing, irrigation and reservoir decisions from calibrated odds

ENVIRONMENTAL

Smarter water-resource management through monsoon variability

OPERATIONAL

Plugs onto NCMRWF’s existing NWP output — no new weather model

SCIENTIFIC

Regime-wise error diagnostics, e.g. raw GEFS beats ML correctors in depressions

TECHNOLOGICAL

Offline, explainable, auditable — every number traces to data + code

Who benefits



IMD / NCMRWF forecasters



NDMA / SDMA disaster mgmt



Farmers & agri planners



Reservoir & water managers



Citizens in flood-prone areas

Key research findings (validated, then held-out)



Regime-awareness helps — modestly and provably

Beats the regime-blind model on RMSE in validation and on both held-out sets (test: 0.08 mm, 95% CI 0.03–0.14); strongest over the Western Ghats



Deep learning wins on heavy rain

U-Net heavy-rain ETS 0.155 vs 0.098 (test) and 0.096 vs 0.070 (live forecasts), both significant; wetter on average, so it is an extra layer



Heavy rain needs probabilities; splitting data hurts

Yes/no exceedance scores below climatology; our probability model AUC 0.90 on test years; per-regime quantile mapping stays worse even with 16 seasons

References (click to open)

[Rajeevan, Gadgil & Bhate \(2010\) — Active and break spells of the Indian summer monsoon, J. Earth Syst. Sci.](#)
[Pai et al. \(2014\) — IMD 0.25-degree gridded daily rainfall, MAUSAM](#)
[NOAA GEFSv12 Reforecast — AWS Open Data Registry](#)
[Roberts & Lean \(2008\) — Fractions Skill Score, Monthly Weather Review](#)
[IMD RSMC New Delhi — cyclone and monsoon-depression best track](#)
[Hunt — ERA5 South Asian low-pressure-system catalogue \(Zenodo, CC BY 4.0\)](#)
[District boundaries — DataMeet / Survey of India index maps](#)

Comparison with existing approaches — day 1, unseen 2018–19

Metric	Raw NWP	Quantile mapping	Global ML	Regime-aware C1	Regime-split C2	U-Net (DL)
RMSE (mm) ↓	16.73	17.99	13.49	13.39	13.47	13.51
Correlation ↑	0.487	0.496	0.599	0.607	0.601	0.613
ETS ≥64.5 mm ↑	0.108	0.151	0.099	0.098	0.100	0.155
FSS ≥64.5 mm ↑	0.520	0.593	0.357	0.353	0.365	0.551

Test years 2018–19: never used for training or tuning, scored once. Green = best. ML models cut average error; the U-Net and quantile mapping keep heavy-rain skill — heavy rain is also issued as a probability.

What sets RegimeCast apart

Capability	Raw NWP	Single bias correction	RegimeCast
Corrects systematic NWP bias	✗	✓	✓
Adapts correction to weather regime	✗	✗	✓
Calibrated heavy-rain probability	✗	✗	✓
Explanation shipped with each forecast	✗	✗	✓